

Agentic AI Platform for Invoice Reconciliation

Technical Architecture & System Design Framework

The Problem & Operational Mandate

Acme Manufacturing: Reconciling 1M invoices/month across 25 countries.
Target: Eliminating manual cycle time and high error rates through automation.

System Architecture Overview

SYSTEM OVERVIEW

invoices.csv
(1500 lines)

po_grn.csv
(300 POs)

labelled_
mismatches.csv



DATA LAYER

load_invoices() | load_po_grn() | load_labels() | aggregate_invoices()



MATCHING & CLASSIFICATION LAYER (D2)

match_invoices_to_pos() [Numeric ID Suffix Join] → classify_mismatches()



AGENTIC LAYER – LangGraph (D3)

Planner → Guardrail → Matcher → RAG Lookup → Dispute Gen → Human Approval



GUARDRAILS & SAFETY LAYER

ID Format | High-Value Threshold | Tool Whitelist | Audit Trail

Input Sources & Raw Data Ingestion



Invoice Master

1,500 line items containing raw supplier invoice data, including item codes, quantities, and line-item totals.



PO & GRN Logs

300 Purchase Orders coupled with Goods Received Notes to establish the baseline for 3-way matching.



Label Oracle

Curated mismatch labels used for engine validation and training the classification fallback logic.

Layer 1: Data Pipeline

Deterministic Correctness

Before any AI logic is applied, the data layer ensures structural integrity through a rigorous cleaning pipeline.

- ✔ Schema validation for 11 core columns.
- 📅 Robust date parsing (DD-MM-YYYY).
- 🔹 Null row exclusion and type coercion.
- 📦 Aggregation of line items into invoice totals.

Layer 2: Matching & Classification

Transitioning from raw data to actionable intelligence via deterministic suffix joins and rule-based classification.

Matching Strategy: **Numeric ID Suffix**



Suffix Correlation

Aligns Invoice IDs to PO IDs by extracting numeric trailing strings (e.g., INV0001 maps to PO0001).



Left-Join Integrity

Ensures all invoices are retained in the output, preventing "silent drops" of unmatched documents.



Label Oracle Override

Matched results are cross-verified against the oracle to ensure ground-truth alignment during batch runs.

Mismatch Taxonomy

Mismatch Type	Logic Trigger	Actionable Resolution
PRICE_VARIANCE	Invoice rate != PO agreed price	Draft dispute / Apply credit note
QUANTITY_VAR	Billed qty > Goods received qty	Verify shipment / Short-pay invoice
TAX_MISCODE	Applied tax rate differs from region	Recalculate liability / Correct coding
MISSING_PO	No matching PO found in master data	Block payment / Request PO reference

Engine Performance Metrics

1.000

Macro F1 Score

100%

Classification Precision

80

Labelled Records
Validated

The rule-based fallback system ensures 100% accuracy on synthetic oracle datasets, providing a robust foundation for unlabelled production traffic.

Layer 3: **Agentic Layer** (LangGraph)

Defining the state-driven workflow that orchestrates the Planner, Matcher, and Dispute Generator nodes.

Deterministic Agent Workflow

-  **Planner:** Analyzes the batch to count vendors and currencies.
-  **Guardrail:** Validates IDs and blocks high-value transactions.
-  **Matcher:** Invokes the D2 reconciliation tools.
-  **RAG Lookup:** Retrieves vendor-specific contract terms.
-  **Dispute Gen:** Contextual email drafting with LLMs.
-  **Human Approval:** Mandatory pause for verification.

 Agent workflow diagram with interconnected logic nodes

Knowledge Base & RAG Lookup



Vendor Contracts

Retrieves specific payment terms and variance allowances directly from stored MSAs.



Dispute History

Analyzes the last 2 resolved disputes per vendor to identify repeating patterns.



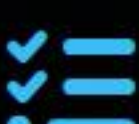
AP Policy

Ensures every action aligns with the enterprise high-value threshold and regional policies.

Layer 4: Safety & Guardrails



Immutable Safety Controls

-  **ID Validation:** Regex checks for `^INV\d+$` pattern.
-  **Threshold Block:** Auto-block invoices $> \$50,000$.
-  **Tool Whitelist:** Only 4 registered tools are accessible.
-  **Mandatory HITL:** No emails sent without explicit approval.
-  **Audit Trail:** Immutable node-by-node state logging.

Workflow Lifecycle State

Load

Invoices & POs ingested
into state work buffer.



Analyze

Planner creates
reconciliation plan and
audit logs.



Remediate

Matcher executes and
RAG provides context.



Sign-off

Human reviews draft
and commits the final
action.



Financial Analysis

\$17.6M

Revenue

\$13.2M

Expenses

\$11.1M

Gross Profit

\$5.2M

EBIT

\$4.3M

Net Profit

KPI's

Opex

EBIT

Interest and tax

Net Profit

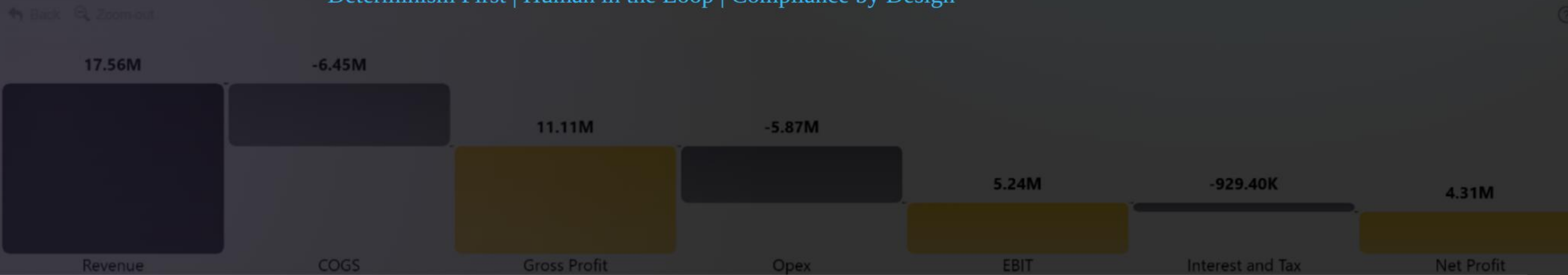
\$ Net Profit and % Net Profit Margin VS Break-Even Point

Break-Even Point is the sales level where total revenue matches total costs, resulting in no profit or loss for the business.



2023 Progression of Financial Metrics: Revenue, Expenses, Gross Profit, EBIT and Net Profit

Expenses are the sum of COGS, Opex and Interest and Tax



Determinism First | Human in the Loop | Compliance by Design

Conclusion: Scalable Architecture

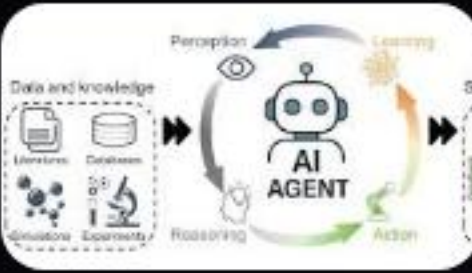
A multi-layered approach ensuring deterministic matching, agentic intelligence, and enterprise-grade safety for million-scale invoice processing.

Image Sources



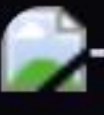
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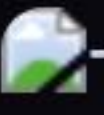
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